

Searches for Shapiro delay in seven binary pulsars using the MeerKAT telescope

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ABSTRACT

Precision timing of millisecond pulsars (MSPs) in binary systems enables observers to detect the relativistic Shapiro delay induced by space–time curvature. When favourably aligned, this enables constraints to be placed on the component masses and system orientation. Here, we present the results of timing campaigns on seven binary MSPs observed with the 64-antenna MeerKAT radio telescope that show evidence of Shapiro delay: PSRs J0101–6422, J1101–6424, J1125–6014, J1514–4946, J1614–2230, J1732–5049, and J1909–3744. Evidence for Shapiro delay was found in all of the systems, and for three the orientations and data quality enabled strong constraints on their orbital inclinations and component masses. For PSRs J1125–6014, J1614–2230, and J1909–3744, we determined pulsar masses to be $M_p = 1.68 \pm 0.17$, 1.94 ± 0.03 , and $1.45 \pm 0.03 M_\odot$, and companion masses to be $M_c = 0.33 \pm 0.02$, 0.495 ± 0.005 , and $0.205 \pm 0.003 M_\odot$, respectively. This provides the first independent confirmation of PSR J1614–2230’s mass, one of the highest known. The Shapiro delays measured for PSRs J0101–6422, J1101–6424, J1514–4946, and J1732–5049 were only weak, and could not provide interesting component mass limits. Despite a large number of MSPs being routinely timed, relatively few have accurate masses via Shapiro delays. We use simulations to show that this is expected, and provide a formula for observers to assess how accurately a pulsar mass can be determined. We also discuss the observed correlation between pulsar companion masses and spin period, and the anticorrelation between recycled pulsar mass and their companion masses.

Key words: telescopes – stars: neutron – pulsars: general – pulsars: individual: PSR J0101–6422, PSR J1101–6424, PSR J1125–6014, PSR J1514–4946, PSR J1614–2230, PSR J1732–5049, PSR J1909–3744.

1 INTRODUCTION

Radio pulsars are rapidly rotating neutron stars, with radio emission arriving at our observatories as a highly predictable train of pulses. The group of pulsars that have a spin period of $P \leq 30$ ms and a rate of spin-down of $\dot{P} \leq 10^{-17}$ are often called millisecond pulsars (MSPs). This subclass is thought to be produced through mass transfer from an evolving giant companion star [during the Roche lobe overflow (RLO) stage] onto the neutron star. The consequent transfer of angular momentum during this stage decreases the spin period of the pulsar, reduces the magnetic field strength and circularizes its orbit (Alpar et al. 1982; Radhakrishnan & Srinivasan 1982; Lorimer 2008; Tauris & van den Heuvel 2023).

Owing to their high rotational stability, MSPs have been used as astrophysical tools in pulsar timing arrays (PTAs) to search for ultra-low (nanohertz) frequency gravitational waves (Shannon et al. 2015; Hobbs & Dai 2017; Arzoumanian et al. 2020; Goncharov et al. 2021; Antoniadis et al. 2022; Chalmers et al. 2022). In some binary pulsars, we can detect relativistic effects in the orbital motion and the propagation of light in the space–time of the system, these are quantified by the so-called ‘post-Keplerian’ (PK) parameters.

With two PK parameters, we can use a theory of gravity to derive the masses of the pulsar and the companion object, as well as the orbital inclination angle with respect to our line of sight (Kiziltan et al. 2013; Özel & Freire 2016). Measuring pulsar masses helps us to determine the underlying distribution of neutron star masses (see also Antoniadis et al. 2016). The neutron stars with highest masses probe the equation of state (EOS) of nuclear matter (Lattimer & Prakash 2004) (for instance, PSR J1614–2230 Arzoumanian et al. 2018,

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PSR J0348+0432 Antoniadis et al. 2013, and PSR J0740+6620 Fonseca et al. 2021), since any EOS that fails to predict such massive neutron stars can be ruled out (Lattimer & Prakash 2004; Özel & Freire 2016).¹

If we measure more than two PK parameters, we can test the theory of gravity being used to interpret them, by verifying whether each pair of PK parameters yields consistent mass estimates (Stairs 2003; Kramer et al. 2006; Kramer 2014; Krishnan 2019; Kramer et al. 2021a; Freire 2022). In some extreme cases, we can even look for other relativistic effects like spin–orbit coupling (Weisberg, Romani & Taylor 1989; Stairs, Thorsett & Arzoumanian 2004; Fonseca, Stairs & Thorsett 2014; Venkatraman Krishnan et al. 2020), which might, in some cases, provide additional tests of GR and even constrain the neutron star EOS (Hu et al. 2020).

In pulsar timing, it is a standard practice to develop a timing model via an iterative procedure that can accurately predict the time of arrival of every pulse (ToA). The timing model includes parameters that describe the pulsar’s spin, astrometric, and orbital properties; however, there are also noise processes that can be correlated with parameters in the timing model. The timing model is then iteratively updated using software such as TEMPO2 (Edwards, Hobbs & Manchester 2006; Hobbs, Edwards & Manchester 2006), which is used to calculate and analyse timing residuals. To properly analyse the arrival times, we need to account for the noise processes, which for the purposes of this analysis, are limited to the systematic white noise, achromatic red noise, and that induced by dispersion measure variations.

According to the general theory of relativity, the Shapiro delay is the additional delay in the propagation of any signal travelling at the speed of light caused by the space–time curvature near massive objects (Shapiro 1964). If a pulsar’s orbit is observed close to edge-on, the Shapiro delay caused by the space–time curvature near the companion can be detected in the pulsar signal; in such cases, it can be used to constrain the component masses and system’s orbital inclination. The shape of the timing residuals as a function of orbital phase determines the orientation, and the amplitude of the delay peak is proportional to the companion mass.

In this paper, we present timing and noise analyses of seven binary MSPs: PSR J0101–6422, J1101–6424, J1125–6014, J1514–4946, J1614–2230, J1732–5049, and J1909–3744, using observations from the 64-antenna MeerKAT radio telescope.

Among these MSPs, PSRs J1909–3744 and J1614–2230 are of great importance. PSR J1909–3744 has an extremely narrow pulse (Jacoby et al. 2003). For that reason, it has the highest timing precision of any pulsars known (Miles et al. 2022). This, in combination with the high orbital inclination resulted in the first precise mass measurement for an MSP (Jacoby et al. 2005). The high timing precision means it has been timed by all the major timing arrays (Shannon et al. 2013; Lentati et al. 2015; Arzoumanian et al. 2020); this has resulted in multiple independent measurements of its mass (Arzoumanian et al. 2018; Liu et al. 2020; Reardon et al. 2021).

For more than a decade, PSR J1614–2230 has been one of the most important pulsars for constraining EOSs due to its high mass. Demorest et al. (2010) found the mass of the pulsar to be $1.97 \pm 0.04 M_{\odot}$ by measuring the amplitude of the Shapiro time delay with the Green Bank Telescope. This mass has been refined since (Fonseca

et al. 2016; Arzoumanian et al. 2018), but these masses have never been confirmed with data from other radio telescopes.

The MeerTime Pulsar Timing Array (MPTA) has been established to frequently observe up to 89 pulsars with extremely high precision (Miles et al. 2022) with the aims of searching for gravitational waves coming from merging supermassive black-hole binaries, testing theories of gravity, studying the structure of interstellar plasma, exploring the interior structure of neutron stars, and understanding the origin and evolution of MSP binary systems.² By augmenting the MPTA data set with intense observing campaigns around superior conjunction, the ongoing Relativistic Binary programme (Kramer et al. 2021b) under the MeerTime Large Survey Project with the MeerKAT telescope (Bailes et al. 2020) is aiming to increase the number of known pulsar masses amongst other objectives (e.g. Serylak et al. 2022). This paper, as a part of the Relativistic Binary programme, investigates measurements of Shapiro delay and (where possible) the derived masses of seven MSPs and their companion stars. In Section 2, we describe the MeerKAT observations and the data processing leading to the calculation of the ToAs. In Section 3, we present our methods for modelling noise processes, updating the timing models, and measuring the Shapiro delays. In Section 4, we present our results on constrained masses and orbital inclination angles. We also present two simulations performed for exploring the change in the pulsar mass error as a function of the orbital inclination angle. In Section 5, we discuss our measurements and compare them to previous values, finding good agreement and explore correlations between the component masses and the spin periods of recycled pulsars. In Section 6, we summarize the major results and draw our conclusions.

2 THE DATA SET

The observations of the MSP sample began in early 2019 and concluded in early 2022. Observations were also conducted as part of the MPTA project (Miles et al. 2022, in press). The observations were performed with the 64-dish MeerKAT radio telescope using two receivers: the ultra-high Frequency (UHF) receiver, operating between 544 and 1088 MHz, and the *L*-band receiver, operating between 856 and 1712 MHz, both with 1024 frequency channels (Bailes et al. 2020). Using the two receivers, we collected coherently de-dispersed data folded in real time at the apparent pulsar period; at each observing session only one receiver was used. PSRs J1101–6424, J1125–6014, J1514–4946, J1614–2230, and J1909–3744 have been observed with the *L*-band receiver only. UHF observations of PSR J0101–6422 and J1732–5049 were included as these had relatively steep spectral indices of -1.7 and -1.9 , respectively (Gitika et al., in preparation) and benefitted from UHF observing campaigns near superior conjunction. The minimum number of 37 observations was for PSR J1514–4946 and the maximum number of 187 observations was for J1909–3744, over a three-year time-span, and have good orbital coverage. For each pulsar, we conducted a long, targeted observing campaign, centred in time near superior conjunction (where the companion passes between the Earth and the pulsar), to optimize our sensitivity to the Shapiro delay. A summary of the data spans, total integration times, and the number of observations for each pulsar are listed in Tables A1 and A2.

Frequency channels affected with radio–frequency interference (RFI) were zero-weighted with the RFIHUNTER RFI rejection tool.³

¹We note that even more massive neutron stars could exist (e.g. Romani et al. 2021, 2022), but such masses were not obtained from the measurement of PK parameters, and rely on assumptions about surface heating and reradiation and are more model-dependent as a result.

²www.meertime.org

³<https://github.com/mshamohammadi/rfihunter>

This software subtracts a normalized and phase-aligned mean pulse profile from the data and creates pulse profile residuals. To do this, it fits a smoothed integrated pulse profile to each 8-s integrated frequency channel, using a non-linear least-squares fitting algorithm. The pulse residuals are then searched for the detection of RFI-affected frequency channels, after normalizing with a ‘median band-pass model’. Following this, the frequency channels with statistics above 3.5σ from the median are zero-weighted, similar to the method that was employed in Bailes et al. (2020). The statistics used for RFI excision are (1) the maximum amplitude of the Fourier transform of the mean-subtracted residuals, (2) the standard deviation, and (3) the range (following the statistics used in Lazarus et al. 2016). To construct a median bandpass model of each receiver, we used the off-pulse baselines and used the Gaussian Process Regression module⁴ implemented in SCIKIT-LEARN package (Pedregosa et al. 2011) to determine a robust median.

After RFI excision, we installed pulsar ephemerides updated for the work of Spiewak et al. (2022, private communication), averaged the observations in time and reduced the data to 16 frequency channels. This channelization allowed us to take into account the temporal variations of pulsar dispersion-measure (DM) (Keith et al. 2013; Jones et al. 2017), as well as the detection of the in-band profile evolution, which requires frequency-dependent parameters in our timing solution (Demorest et al. 2013).

For every pulsar, we summed high-signal-to-noise ratio observations in time, polarization, and frequency, and then using wavelet smoothing techniques implemented in the PSRCHIVE program `psrsmooth`, we created a noise-free pulse profile, often termed a ‘standard’ or ‘template’. For PSRs J1614–2230 and J1909–3744, we used `PulsePortraiture`⁵ to create a frequency-dependent template, which is a profile that evolves in frequency, referred to as a ‘portrait’ (Pennucci 2019), the aim of which is to account for frequency-dependent time delays caused by profile evolution. This was possible because we had very high-signal-to-noise ratio observations of these two pulsars, making the profile evolution with frequency secure.

Finally, we used the Fourier domain Monte Carlo algorithm in `pat`, which fits for a phase gradient between the Fourier transforms of the template and profile, to calculate a ToA for every pulse profile, and selected the profiles with a signal-to-noise ratio higher than 10. This algorithm uses a Markov chain Monte Carlo method to determine the likely ToA uncertainties.⁶

3 METHODS

For standard pulsar timing practices, we need a timing model (solution) that includes pulsar parameters such as its astrometry, spin, dispersion measure, binary motion (five Keplerian parameters), and a Solar system model. The astrometric parameters include position (right ascension α and declination δ), proper motion ($\mu_\alpha \cos \delta$, μ_δ), and when significant, a parallax. When profiles possess strong profile evolution as a function of frequency, radio frequency derivative parameters are added to the timing model. A list of systematic jumps are also added to the timing model if needed between frequency bands or observational set-ups. We commenced our timing with the latest pulsar ephemerides from Spiewak et al. (2022, private communication) as the initial timing models, and added two Shapiro

parameters (will be discussed below) into the models, and refined these based upon our ToAs using TEMPO2, utilizing a generalized least-squares fitting method to minimize the timing residuals, and updated them constantly by adding ToAs from new observations. We used JPL DE440 as our Solar system ephemeris (Park et al. 2021), the Bureau International des Poids et Mesures BIPM2020 as our realization of terrestrial time, and Barycentric Coordinate Time (TCB) for our units.

3.1 Timing model

For a pulsar in a binary with a low eccentricity, e , it is difficult to measure the longitude of periastron, ω , and there is a high correlation between the longitude of periastron and the time of periastron, T_0 . In such systems, there is typically a high correlation between the orbital period, P_b , and the time-derivative of the longitude of periastron, $\dot{\omega}$. The ELL1 binary model can address this problem in systems with a low e (Lange et al. 2001). Instead of e , ω , and T_0 , this model uses the time of passage through the ascending node, T_{asc} , and the two Laplace–Lagrange parameters, $\eta \equiv e \sin \omega$ and $\kappa \equiv e \cos \omega$, which can be measured with less covariance.

The effects of Shapiro delay have been parameterized in two binary models implemented in TEMPO2: DD (Damour & Deruelle 1986) and ELL1 using a function of two PK parameters: range (r) and shape (s) in the following forms:

$$r = T_\odot M_c, \quad (1)$$

$$s = \sin i, \quad (2)$$

where M_c is the mass of the companion and $T_\odot \equiv G M_\odot c^{-3} = 4.925490947 \mu\text{s}$ is the mass of the Sun in time units. The Shapiro delay, Δ_s , in the ELL1 binary model is modelled as

$$\Delta_s = -2r \ln(1 - s \sin \Phi), \quad (3)$$

where Φ is the celestial longitude of the binary. At high inclination angles, where $\sin i \sim 1$, the peak of the Shapiro delay is very pronounced, but at other inclinations it becomes extremely covariant with the Römer delay (caused by the pulsar’s orbital motion, Lange et al. 2001) from the circular orbit.

The problem with r and s is that they become highly correlated, in particular at low (i.e. face-on) orbital inclinations. Because of this, Freire & Wex (2010) parameterized the Shapiro delay for low-eccentric binary pulsars in Fourier space, using the sum of the harmonics of the Shapiro delay. They introduced the two new PK parameters that are proportional to the third and fourth harmonic amplitudes (h_3, h_4). For higher inclinations, the best parameterization is h_3 and $\varsigma = h_4/h_3$, because the latter parameter can be measured more precisely than h_4 . The main advantage of these ‘orthometric’ parameters is that they are less correlated with one another compared to r and s , thus providing an optimal description of the $m_c, \sin i$ constraints provided by the Shapiro delay. The orthometric parameters relate to r, s according to the following relations (Freire & Wex 2010):

$$\varsigma = \frac{s}{1 + \sqrt{1 - s^2}}, \quad (4)$$

$$h_3 = r \varsigma^3, \quad (5)$$

and obtained the equivalent Shapiro delay formula

$$\Delta_s = -\frac{2h_3}{\varsigma^3} \ln(1 + \varsigma^2 - 2\varsigma \sin \Phi). \quad (6)$$

⁴https://scikit-learn.org/stable/modules/gaussian_process.html

⁵<https://github.com/pennucci/PulsePortraiture/tree/py3>

⁶The algorithm is in the `ProfileShiftFit.C` source code from the `psrchive` sourceforge.net repository.

Within the range of possible orbital inclinations ($0^\circ \leq i \leq 180^\circ$), we have $0 \leq \sin i \leq 1$; this implies that $0 \leq \varsigma \leq 1$ and $h_3 > 0$ (since $r \equiv T_\odot M_c$ must be positive).

Three new binary models were established based upon the (h_3, h_4) and (h_3, ς) parameterizations and implemented in the ELL1H binary model as an alternative to the ELL1 binary model. In this work, we have used these models with the h_3 and ς parameterization when searching for the relativistic Shapiro delay in the timing residuals for each pulsar because the binaries being studied have relatively high orbital inclinations. We note that, although physical constraints require that $0 \leq \varsigma \leq 1$ and $h_3 > 0$, such constraints are not enforced by the ELL1H orbital model, in the same way that the conditions $0 \leq s \leq 1$ and $r > 0$ are not enforced by any orbital model when fitting for r and s .

3.2 Noise models

There are two basic contributions in calculating timing residuals: the deterministic and stochastic noise components (Lentati et al. 2014). The latter can be modelled using three white and two red stochastic processes:

(i) White (time uncorrelated) noise: can be described using three components as follows:

(a) EFAC: a factor α_i that is multiplied to any ToA uncertainty σ_i , where i is the i th ToA. One possible contribution to this term is due to the radiometer noise after measuring ToA uncertainty using the template-matching algorithm. For every receiver, backend, or pulsar, we might need to use different EFAC parameter.

(b) EQUAD: an error term β_i that is added in quadrature to the scaled ToA uncertainty by EFAC. It is usually necessary to have separate EQUAD parameters for different the receiver, backend, or pulsar. Therefore, the uncertainty of the i th ToA scaled with EFAC and EQUAD can be written as

$$\sigma_{i, \text{whitened}}^2 = (\alpha_i \sigma_i)^2 + \beta_i^2. \quad (7)$$

(c) ECORR: a noise term that describes the correlations between the sub-bands of one observation, and is not correlated between observations taken at different epochs. We used the TECORR parameter described in equation (1) in Bailes et al. (2020) in our noise analysis.

(ii) Red noise: this term accounts an achromatic stationary Gaussian process with a low frequency, and is modelled using a power-law spectrum that has a form of

$$P_{\text{red}}(f) = A_{\text{red}}^2 \left(\frac{f}{1 \text{ yr}^{-1}} \right)^{\gamma_{\text{red}}}, \quad (8)$$

where $S_{\text{red}}(f)$ is the red noise power spectral density at frequency f . A_{red} is the amplitude of the red noise in $\mu\text{s yr}^{1/2}$, and γ_{red} is the spectral index of the noise process.

(iii) Dispersion noise: pulses at higher frequencies arrive earlier in a dispersive interstellar medium (ISM). The delay due to the frequency dispersion is measured (in seconds) using $1/2.41 \times 10^{-4} DM/\nu^2$, where DM is the dispersion measure and ν is the observing frequency in MHz. DM , measured in pc cm^{-3} , is the integral of the electron number density along our line of sight

$$DM = \int_0^L n_e(z) dz, \quad (9)$$

where L is the distance to the pulsar. Our line of sight changes due to the motion of the pulsar and the Earth, causing changes in the DM with time. Additionally, the turbulent ISM between the pulsar and the Earth has a bulk motion, which induces additional changes in the DM with time. Therefore, DM is epoch dependent, and DM corrections are required after every observation. Through the use of broad-band observations, it is possible to mitigate the effects of dispersion measure variations.

This temporal DM variation introduces an additional noise term in the timing residuals. We describe this temporal variation using a power-law spectrum that has a form similar to the red noise spectrum,

$$P_{\text{DM}}(f) = A_{\text{DM}}^2 \left(\frac{f}{1 \text{ yr}^{-1}} \right)^{\gamma_{\text{DM}}} \left(\frac{\nu}{\nu_c} \right)^{-2}, \quad (10)$$

where A_{DM} is the DM noise amplitude that scales with ν^{-2} , and ν_c is the centre frequency of the observation.

3.3 Noise analysis

We perform our noise analysis with TEMPONEST (Lentati et al. 2014), utilizing an efficient Bayesian inference sampler, Multinest (Feroz & Hobson 2008; Feroz, Hobson & Bridges 2009), for exploring the parameter space and producing marginal posterior distributions of parameters. Model comparison between two timing models can be quantified using the ratio of their evidence, which is called the Bayes factor $\mathcal{B}$. Using the evidences returned by TEMPONEST, we calculate the log Bayes factor, $\ln \mathcal{B}$, and define significant support for one model in comparison to another model as $\mathcal{B} \geq 10$ (Kass & Raftery 1995), or $\ln \mathcal{B} \geq 2.3$.

3.4 Relativistic parameter derivations

There are additional physical parameters that can be derived from the measured parameters in the pulsar ephemerides. We can constrain the pulsar mass, M_p , the companion mass, M_c , and the orbital inclination angle, i , by measuring the relativistic Shapiro delay parameters, h_3 and ς , in our timing residuals. Every binary system has a precise binary mass function, f_m :

$$f_m = \frac{(M_c \sin i)^3}{(M_p + M_c)^2} = \frac{4\pi^2 (a_p \sin i)^3}{G P_b^2} = \frac{4\pi^2 x^3}{T_\odot P_b^2}, \quad (11)$$

which relates the unknown masses and the orbital inclination angle of the system. The mass function can be calculated using the observed orbital period and the observed projected semimajor axis of the pulsar, $x = a_p \sin i/c$ in units of light-seconds, where a_p is the semimajor axis of the pulsar, and c is the speed of light. The quantity x can also experience a secular change, and it can experience a periodic modulation due to the orbital motion of the Earth around the Sun. This annual motion of the Earth causes the observed orbital inclination angle to have an annual periodic change that can be obtained by measuring the annual orbital parallax. In the case of the apparent change in x due to the proper motion, from Kopeikin (1996), the maximum value of $\dot{x}/x$ provides an upper limit of

$$i \leq \tan^{-1} \left[1.54 \times 10^{-16} \left(\frac{\mu_T}{\text{mas yr}^{-1}} \right) / \left(\frac{\dot{x}}{x} \right)_{\text{max}} \right] \quad (12)$$

for the orbital inclination angle, where μ_T is the total proper motion of the system in mas yr^{-1} . This provides another way to constrain the system inclination angle using the $\dot{x}$ measurement.

Table 1. Pulsar and companion star masses and inclination angles. The derived inclination angles have another equally likely solution of $180^\circ - i$. The pulsars with two entries could not have their pulsar masses well determined and so two possible (small and large) pulsar masses were held fixed and likely companion and inclination angles provided for them.

Pulsar	M_p ($M_\odot$)	M_c ($M_\odot$)	$\cos i$	i°
J0101–6422	1.2	0.14–0.16	0.24–0.28	73–75
	2.0	0.20–0.22	0.29–0.34	70–74
J1101–6424	1.2	0.51–0.59	0.47–0.60	53–62
	2.0	0.74–0.87	0.55–0.67	47–57
J1125–6014	$1.68^{+0.17}_{-0.15}$	$0.33^{+0.02}_{-0.02}$	$0.215^{+0.013}_{-0.012}$	$77.6^{+0.7}_{-0.8}$
J1514–4946	1.2	0.15–0.17	0.14–0.28	73–82
	2.0	0.22–0.24	0.21–0.36	68–78
J1614–2230	$1.94^{+0.03}_{-0.03}$	$0.495^{+0.005}_{-0.005}$	$0.0143^{+0.0002}_{-0.0002}$	$89.179^{+0.013}_{-0.013}$
J1732–5049	1.2	0.17–0.20	0.31–0.47	61–72
	2.0	0.24–0.27	0.35–0.51	59–70
J1909–3744	$1.45^{+0.03}_{-0.03}$	$0.205^{+0.003}_{-0.003}$	$0.0573^{+0.0016}_{-0.0016}$	$86.69^{+0.10}_{-0.10}$

In order to constrain the pulsar mass through the measurement of the Shapiro delay, it is necessary to constrain both the companion mass and the system inclination angle. So, the error in the pulsar mass σ_{M_p} , depends on the errors in both the companion mass σ_{M_c} and the sine of the inclination angle $\sigma_{\sin i}$, and can be estimated using partial derivatives of the mass function (equation 11),

$$\sigma_{M_p} \approx \sqrt{A^2 \sigma_{M_c}^2 + B^2 \sigma_{\sin i}^2 + 2AB \text{Cov}[M_c, \sin i]} \quad (13)$$

,where

$$A = \frac{3M_c^{1/2} \sin i^{3/2}}{2f_m^{1/2}} - 1, \quad (14)$$

$$B = \frac{3M_c^{3/2} \sin i^{1/2}}{2f_m^{1/2}}, \quad (15)$$

and $\text{Cov}[M_c, \sin i]$ is the covariance between M_c and $\sin i$. We note that this equation was used for an efficient estimate of the uncertainties of M_p in the simulations, not to estimate the uncertainties of M_p for the pulsars in his work.

4 RESULTS

A summary of spin, astrometric, orbital, and noise parameters for each pulsar is listed in Tables A1 and A2 in Appendix A. For PSRs J1125–6014, J1614–2230, and J1909–3744, we report measurements of orthometric Shapiro parameters of h_3 and ζ , and from these we have derived pulsar masses, the companion masses, and the orbital inclination angles. Figs C1–C3 provide the two-dimensional posteriors of the measured and the derived parameters. Among the seven pulsars, we have significantly measured the pulsar and the companion star masses for PSR J1125–6014, J1614–2230, and J1909–3744. These are described in Section 4.1 and summarized in Table 1. Considering the posterior probability distributions of the derived parameters, the values and their uncertainties are taken from the cumulative distribution function at the values of 0.16 (lower bound), 0.5 (median), and 0.84 (upper bound). We define the orbital inclination angles of the MSP binary systems to be in the range of $0^\circ < i < 180^\circ$. This definition satisfies the standard astronomical convention. In this work, we only report the solutions for the inclinations that are between 0° and 90° , corresponding to face-on and edge-on systems, respectively. However, there are another equally likely solutions of $180 - i$. We also quantify how the error

in pulsar mass scales with $\cos i$ using simulated ToAs in Section 4.2, and explain why we were not able to measure the pulsar mass in the other four systems.

4.1 Measuring the Shapiro delay and constraining pulsar and companion masses

We saw significant evidence for Shapiro delay in all seven systems, but could only accurately measure the pulsar masses in PSRs J1125–6014, J1614–2230, and J1909–3744. For PSRs J0101–6422, J1101–6424, J1514–4946, and J1732–5049, it was not possible for us to constrain the orthometric parameters significantly, as the uncertainties in the pulsar masses were greater than $2 M_\odot$. Instead, for these systems, we assumed the pulsar mass to be greater than $1.2 M_\odot$ and for linearly spaced $\cos i$ values from 0.01 to 0.99, with a resolution of 0.01, we calculated a series of (h_3, ζ) pairs. We then used every pair in the timing model and held them fixed while fitting for other parameters, and recorded a χ^2 value from TEMPO2. Afterwards, we differentiated the χ^2 values from the global minimum, $\chi_{\min}^2$, to find a $\Delta\chi^2$ value for every $\cos i$. Finally, we mapped them to a Bayesian likelihood function (Splaver et al. 2002), and calculated the 1σ standard deviation of $\cos i$. This gave an estimate of the orbital inclination angle, and using the mass function formula (equation 11), we could find the 1σ standard deviation of the companion mass. We repeated this analysis for a heavy pulsar mass of $2.0 M_\odot$ as well. Using the two, we could determine how likely the companion mass and the orbital inclination angle of the systems are, assuming the pulsar mass to be between 1.2 and $2.0 M_\odot$. We believe that the chosen values of 1.2 and $2.0 M_\odot$ are justifiable as almost all known MSPs lie in this range, and we use this as a prior to explore what the companion masses and inclination angles are. These reported upper and lower limits of M_c and i are therefore reasonably conservative. Recently, it has been argued that PSR J0952–0607 has a pulsar mass of $2.35 \pm 0.17 M_\odot$ by examining its luminosity as a function of orbital phase and using its mass ratio from spectroscopic observations of the companion (Romani et al. 2022). If MSPs can be as massive as $\sim 2.35 M_\odot$ this would weaken these companion mass and inclination angle constraints but only mildly.

4.1.1 PSR J0101–6422

PSR J0101–6422 is an MSP with a spin period of 2.57 ms in a 1.78 d orbit. It is an unidentified γ -ray source in the first *Fermi* Large Area Telescope catalogue (Abdo et al. 2010), and was confirmed as an MSP with the Parkes radio telescope, also known as Murriyang, by Kerr et al. (2012). The results from the noise analysis showed that the model containing the white noise and the DM noise parameters had the highest Bayesian evidence. Comparing two-timing models, including one with the Shapiro parameters and one without, resulted in obtaining a log Bayes factor of $\ln \mathcal{B} \sim 21$, demonstrating that the model including the Shapiro parameters is preferred. We detected the Shapiro delay with $\sim 4\sigma$ significance in h_3 and $\sim 8\sigma$ significance in ζ . Using the χ^2 analysis mentioned in 4.1, we place a constraint on $\cos i$ to be in the range of $0.24 \leq \cos i \leq 0.28$ if $M_p = 1.2 M_\odot$, and $0.29 \leq \cos i \leq 0.34$ if $M_p = 2.0 M_\odot$. The corresponding companion mass and orbital inclination angle for the pulsar mass of $M_p = 1.2 M_\odot$ found to be in the ranges of $0.14 \leq M_c \leq 0.16 M_\odot$ and $73^\circ \leq i \leq 75^\circ$, and for the pulsar mass of $M_p = 2.0 M_\odot$ found to be in the ranges of $0.20 M_\odot \leq M_c \leq 0.22 M_\odot$ and $70^\circ \leq i \leq 74^\circ$, respectively. These values are consistent with the predictions of Tauris & Savonije (1999) for a Helium white dwarf (He WD) star. Although these values are

reasonably well bounded, they do not lead to interesting constraints on the pulsar's mass.

4.1.2 PSR J1101–6424

PSR J1101–6424 is an MSP with a spin period of 5.11 ms in a 9.61 d orbit. Ng et al. (2015) discovered the pulsar, during processing of the low-latitude Galactic plane data sets in the High Time Resolution Universe pulsar survey. Comparing the noise models, the model with the white noise and the DM noise parameters had the highest Bayesian evidence. We performed the noise analyses using two-timing models with and without the Shapiro delay parameters, and obtained a log Bayes factor of $\ln \mathcal{B} \sim 5$. This indicates that we have a weak detection of Shapiro delay in our timing residuals. Using the χ^2 analysis mentioned in 4.1, we place a constraint on $\cos i$ to be in the range of $0.47 \leq \cos i \leq 0.60$ if $M_p = 1.2 M_\odot$, and $0.55 \leq \cos i \leq 0.67$ if $M_p = 2.0 M_\odot$. The corresponding companion mass and orbital inclination angle for the pulsar mass of $M_p = 1.2 M_\odot$ found to be in the ranges of $0.51 \leq M_c \leq 0.59 M_\odot$ and $53^\circ \leq i \leq 62^\circ$, and for the pulsar mass of $M_p = 2.0 M_\odot$ found to be in the ranges of $0.74 \leq M_c \leq 0.87 M_\odot$ and $47^\circ \leq i \leq 57^\circ$, respectively. This is an interesting system, as the spin period (and even the orbital period) would suggest it is similar to J1614–2230, meaning that is possibly originated as Case A RLO (see discussion in section 5). The value of M_c might be consistent with that of J1614–2230.

4.1.3 PSR J1125–6014

PSR J1125–6014 is an MSP with a spin period of 2.63 ms in a 8.75 d orbit. It is one of the 15 MSPs discovered by Faulkner et al. (2004) through a reprocessing of the Parkes Multibeam Pulsar Survey. The results from performing noise analysis on the PSR J1125–6014 timing residuals show that the noise model with the white noise and the DM noise parameters had the highest Bayesian evidence. We derive and constrain the mass of the pulsar to be $M_p = 1.68 \pm 0.17 M_\odot$. The companion mass and the orbital inclination angle are constrained to be $M_c = 0.33 \pm 0.02 M_\odot$ and $i = 77.6 \pm 0.8^\circ$, respectively. In Fig. C1 of Appendix C, the posterior probability distributions of the measured Shapiro parameters are presented as well as the derived masses and inclination angles for PSR J1125–6014. We discuss the nature of this system in more detail in Section 5.

4.1.4 PSR J1514–4946

PSR J1514–4946 is an MSP with a spin period of 3.59 ms in a 1.92 d orbit. It is an unknown source in the first *Fermi* Large Area Telescope catalogue (Abdo et al. 2010), and was confirmed as an MSP with the Parkes radio telescope by Kerr et al. (2012). Its phase-connected timing solution was later determined by Camilo et al. (2015). We performed a noise analysis on the PSR J1514–4946 timing residuals and obtained the same Bayesian evidences for noise models with DM, red, and white noise parameters. Therefore, we chose the noise model contained only the white noise parameters as it is the simplest one. We obtained a log Bayes factor of $\ln \mathcal{B} \sim 6$ by comparing the timing models with and without the Shapiro parameters, showing that we detected a weak Shapiro signature in the timing residuals. Using the χ^2 analysis aforementioned in 4.1, we place a constraint on $\cos i$ to be in the range of $0.14 \leq \cos i \leq 0.28$ if $M_p = 1.2 M_\odot$, and $0.21 \leq \cos i \leq 0.36$ if $M_p = 2.0 M_\odot$. The corresponding companion mass for the pulsar mass of $M_p = 1.2 M_\odot$ found to be in the ranges of $0.15 \leq M_c \leq 0.17 M_\odot$ and $73^\circ \leq i \leq 82^\circ$, and for the pulsar mass of

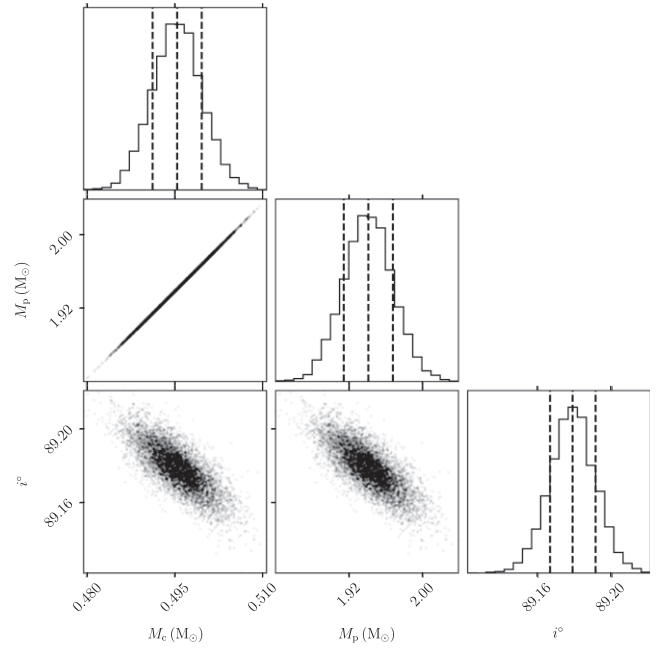


Figure 1. 2D marginalized posterior probability distributions of M_p , M_c , and i (derived from measured h_3 and ζ) for J1614–2230.

$M_p = 2.0 M_\odot$ found to be in the ranges of $0.22 \leq M_c \leq 0.24 M_\odot$ and $68^\circ \leq i \leq 78^\circ$, respectively. These values of M_c are consistent with the Tauris & Savonije (1999) prediction for a He WD companion.

4.1.5 PSR J1614–2230

PSR J1614–2230 is an MSP with a spin period of 3.15 ms in a 8.69 d orbit. Crawford et al. (2006) discovered PSR J1614–2230 after conducting an intermediate-latitude Galactic survey of 56 unidentified γ -ray sources from the third Energetic Gamma-Ray Experiment Telescope catalogue (Hartman et al. 1999) with the 13-beam Parkes multibeam receiver at 1400 MHz (Hessels et al. 2005; Crawford et al. 2006). The precise Shapiro delay measurement, and high mass derived for PSR J1614–2230 carried out by Demorest et al. (2010) was able to reject many of the EOSs of the nuclear matter. Among our pulsars, PSR J1614–2230 has the most significant Shapiro delay detection: $h_3 = 2.340 \pm 0.019 \mu\text{s}$ and $\zeta = 0.9858 \pm 0.0003$. These led us to constrain the mass of the pulsar to be $M_p = 1.94 \pm 0.03 M_\odot$, the companion mass to be $M_c = 0.495 \pm 0.005 M_\odot$, and the orbital inclination angle to be $i = 89.179 \pm 0.013^\circ$. Fig. 1 shows the correlations between these parameters. The timing residuals, pre- and post-fitting, are shown in Fig. 2. The posterior probability distributions of h_3 and ζ , as well as M_c , M_p , and i that are derived from h_3 and ζ , are plotted in Fig. C2 of Appendix C. For PSR J1614–2230, the noise model containing the DM noise, red noise, and white noise parameters had the highest Bayesian evidence.

4.1.6 PSR J1732–5049

PSR J1732–5049 is an MSP with a spin period of 5.31 ms in a 5.26 d orbit. It is one of the nine pulsars discovered by Edwards & Bailes (2001) in an intermediate-latitude Galactic survey with a centre frequency near 1400 MHz conducted with the Parkes radio telescope. From our timing noise analysis, we obtained the highest evidence for the noise model containing the red noise and white

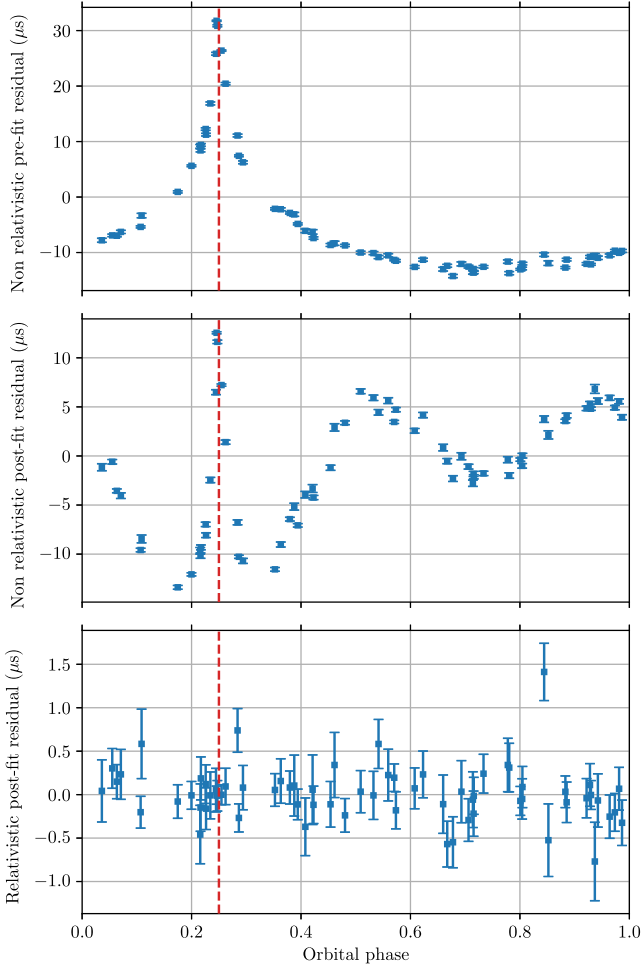


Figure 2. Pre- and post-fit timing residuals as a function of orbital phase for J1614–2230. The top panel shows the actual Shapiro delay signature preserved in the pre-fit timing residuals (excluding relativistic effects from the timing model). The middle panel shows the timing residuals after fitting for the non-relativistic orbital parameters. The bottom panel shows the timing residuals after fitting for all the parameters. The superior conjunction happens at the orbital phase of 0.25 and is indicated by the vertical dashed red line.

noise parameters. Comparing the timing solutions with and without the Shapiro parameters, PSR J1732–5049 showed a likely Bayesian evidence for the presence of a Shapiro signature with a log Bayes factor of $\log \mathcal{B} \sim 6$. Using the χ^2 analysis mentioned in 4.1, we place a constraint on $\cos i$ to be in the range of $0.31 \leq \cos i \leq 0.47$ if $M_p = 1.2 M_\odot$, and $0.35 \leq \cos i \leq 0.51$ if $M_p = 2.0 M_\odot$. The corresponding companion mass for the pulsar mass of $M_p = 1.2 M_\odot$ was found to be in the ranges of $0.17 M_\odot \leq M_c \leq 0.20 M_\odot$ and $61^\circ \leq i \leq 72^\circ$, and for the pulsar mass of $M_p = 2.0 M_\odot$ found to be in the ranges of $0.24 M_\odot \leq M_c \leq 0.27 M_\odot$ and $59^\circ \leq i \leq 70^\circ$, respectively. These values of M_c are consistent with the Tauris & Savonije (1999) prediction for a He WD companion.

4.1.7 PSR J1909–3744

PSR J1909–3744 is an MSP with a spin period of 2.95 ms in a 1.53 d orbit. The Swinburne High Latitude Pulsar Survey was a blind survey that led Jacoby et al. (2003) to discover PSR J1909–3744 using the Parkes 13-beam multibeam receiver. The noise model including DM noise, red noise, and white noise parameters had the highest evidence,

and our constraints on the orthometric Shapiro parameters were $h_3 = 0.845 \pm 0.007 \mu\text{s}$ and $\zeta = 0.9441 \pm 0.0016$. We derive and constrain the mass of the pulsar to be $M_p = 1.45 \pm 0.03 M_\odot$. The companion mass and the orbital inclination angle are constrained to be $M_c = 0.205 \pm 0.003 M_\odot$ and $i = 86:69 \pm 0:10$, respectively. The posterior probability distribution of the measured and derived parameters can be found in the Fig. C3 of Appendix C. The companion mass is consistent with the predictions of Tauris & Savonije (1999) for a He WD, and is the first confirmation of this mass relation.

4.2 The uncertainty in the pulsar mass as a function of $\cos(i)$

While it is unfortunate that not all pulsar binaries are in edge-on systems, it is not surprising. Here, we explore the $(M_c, \sin i)$ space where meaningful Shapiro delays can be detected through simulation. For an arbitrary binary that we simulate, we wanted to understand what the companion mass error (σ_{M_c}) and $\sin i$ error ($\sigma_{\sin i}$) are by holding the pulsar mass, companion mass, and orbital period fixed, and then use them to understand what the pulsar mass error (σ_{M_p}) would be after the fit.

Using TEMPO2’s fake plugin, we conducted two sets of simulations. In both simulations, we have used ELL1 binary model. In the first, we investigated how σ_{M_c} and $\sigma_{\sin i}$ generally change as a function of $\cos i$. Then we used equations (11) and (13) in order to predict σ_{M_p} . The results of this simulation provides a mathematical relation by which the error in the pulsar mass σ_{M_p} can be estimated.

4.2.1 Constant M_c and M_p

In the first set of simulations, we aimed to perform the analysis of measuring σ_{M_p} as we are rotating the orbit in a way that the inclination angle of the orbit changes with respect to our line of sight. We took our timing solution for J1614–2230, removed the dispersion measure and the noise parameters, and used it to simulate ToAs. Using equation 11, the projected semimajor axis of the pulsar at different inclination angles from 45° to 89° , given the pulsar mass of $2.0 M_\odot$ and the companion mass of $1.0 M_\odot$, were calculated. Then 1300 ToAs with the root mean square (rms) residuals of $0.1 \mu\text{s}$ were simulated (similar to what can be obtained for some MSPs in a reasonable time), and using TEMPONEST we sampled the binary parameters including P_b , T_{asc} , x , ϵ_1 , ϵ_2 , M_c , and $\sin i$ and recorded the values and errors of $\sin i$, M_c , and $\text{Cov}[M_c, \sin i]$. For every inclination angle, the simulation was repeated 10 times. In panel (a) of Fig. 3, the value and the error bar of every point is, respectively, the average and the 1σ standard deviation of $\sigma_{\sin i}$ taken from 10 TEMPONEST runs. The same procedure was used for σ_{M_c} , $\text{Cov}[M_c, \sin i]$, and σ_{M_p} in the panels (a)–(c), respectively. For every TEMPONEST run, σ_{M_p} was calculated using equation (13).

We next derived empirical relationships between $\cos i$ and $\sigma_{\sin i}$, σ_{M_c} , and $\text{Cov}[M_c, \sin i]$. We found that the best-fitting function to the relationships was of the polynomial form

$$\mathcal{F}(\cos i) = \sum_{j=0}^n a_j \cos^j i, \quad (16)$$

where the coefficients of a_j for σ_{M_c} fitting function ($\mathcal{F}_{\sigma_{M_c}}$), $\sigma_{\sin i}$ fitting function ($\mathcal{F}_{\sigma_{\sin i}}$), and $\text{Cov}[M_c, \sin i]$ fitting function ($\mathcal{F}_{\text{Cov}[M_c, \sin i]}$) can be found by minimizing the sum of least squares. The number of coefficients depends on the orbital inclination angle. We found that, for $0.01 \leq \cos i \leq 0.16$, we need at least a third-order polynomial ($n = 3$). By increasing the range of $\cos i$ to be in $0.01 \leq \cos i \leq 0.35$, corresponding to the orbital inclination angle of $70^\circ \leq i < 90^\circ$,

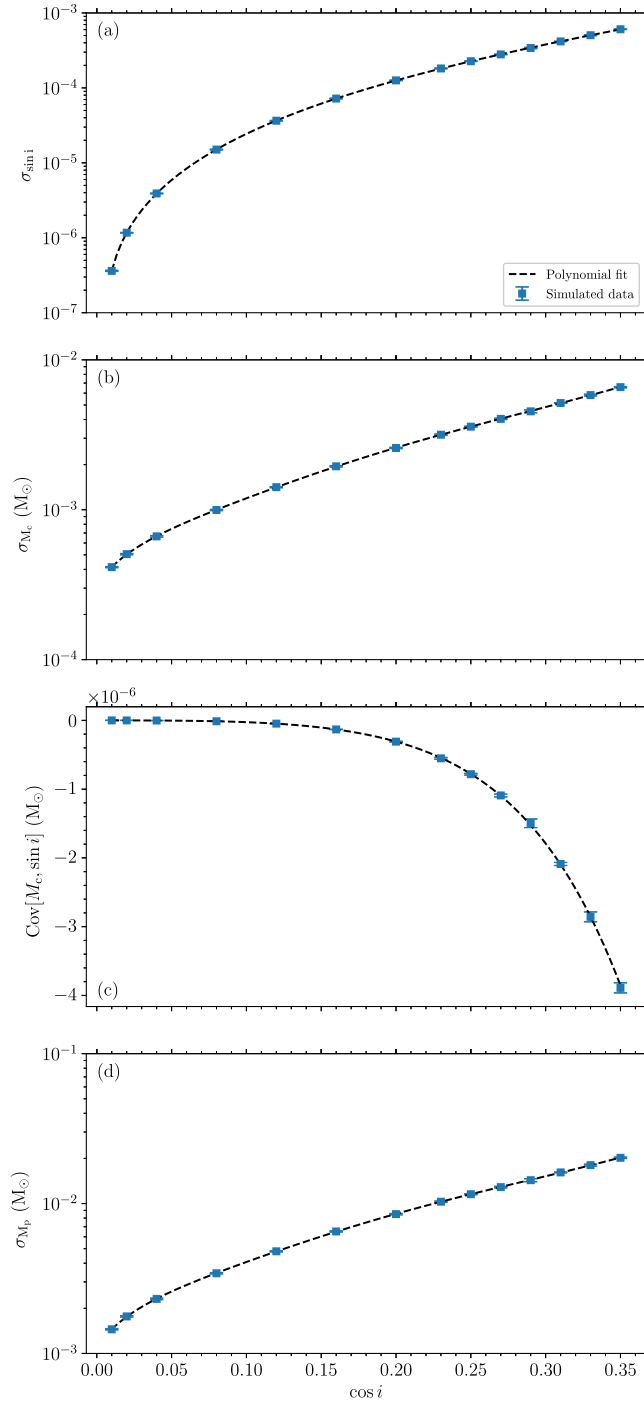


Figure 3. Exploring the error in the component masses at different orbital inclination angles using 1300 simulated ToAs with the rms residuals of $0.1 \mu\text{s}$, for a binary pulsar system with the companion mass of $1 M_\odot$, the pulsar mass of $2.0 M_\odot$, and the orbital period of 8.69 d. Panel (a): the error in the sine of the orbital inclination angle. Panel (b): the error in the companion mass. Panel (c): the covariance between the companion mass and the sine of the orbital inclination angle. Panel (d): the error in the pulsar mass. The dashed lines in panels (a)–(c) are the polynomial functions described by equations (17)–(19), respectively, and the coefficients for each are listed in Table 2. The dashed line in the bottom panel is calculated using equation (13).

Table 2. Fitting results for the error modelling.

Coefficient	$\mathcal{F}_{\sigma_{\sin i}}$	$\mathcal{F}_{\sigma_{M_c}}$	$\mathcal{F}_{\text{Cov}[M_c, \sin i]}$
a_0	$-1.7(4) \times 10^{-6}$	$0.003\,63(10)$	$-9(11) \times 10^{-9}$
a_1	$4.3(6) \times 10^{-4}$	$0.116(9)$	$2(2) \times 10^{-6}$
a_2	$0.014(2)$	$-0.7(2)$	$-2.6(11) \times 10^{-4}$
a_3	$0.10(3)$	$7.0(19)$	$3(3)$
a_4	$0.10(15)$	$-22(7)$	$-0.05(2)$
a_5	$0.4(2)$	$27(8)$	$0.17(9)$
a_6	—	—	$-0.38(13)$

we need to have at least six coefficients. We used the latter case in our modelling of uncertainties, and found the constant coefficients of $a_0, \dots, a_5$ in Table 2 for $\mathcal{F}_{\sigma_{\sin i}}$ (the dashed line in the panel a of Fig. 3), $\mathcal{F}_{\sigma_{M_c}}$ (the dashed line in the panel b of Fig. 3), and $\mathcal{F}_{\text{Cov}[M_c, \sin i]}$ (the dashed line in the panel c of Fig. 3). We then used equation (13) and substituted σ_{M_c} , $\sigma_{\sin i}$, and $\text{Cov}[M_c, \sin i]$ with $\mathcal{F}_{\sigma_{M_c}}$, $\mathcal{F}_{\sigma_{\sin i}}$, and $\mathcal{F}_{\text{Cov}[M_c, \sin i]}$, respectively, to find the modelled error in the pulsar mass (the dashed line in the panel d of Fig. 3). We also explored how $\sigma_{\sin i}$, σ_{M_c} , and $\text{Cov}[M_c, \sin i]$ scale with the rms residuals of 1 and $2 \mu\text{s}$, the companion masses of 0.2 and $0.5 M_\odot$, and with 2600 ToAs (two times the initial number of ToAs). The results show that $\sigma_{\sin i}$, σ_{M_c} , and $\text{Cov}[M_c, \sin i]$ are scaled with the following relations:

$$\sigma_{\sin i} = \left(\frac{M_\odot}{M_c} \right) \left(\frac{\sigma_{\text{rms}}}{\mu\text{s}} \right) \left(\frac{1000}{N_{\text{ToA}}} \right)^{\frac{1}{2}} \mathcal{F}_{\sigma_{\sin i}}(\cos i), \quad (17)$$

$$\sigma_{M_c} = \left(\frac{\sigma_{\text{rms}}}{\mu\text{s}} \right) \left(\frac{1000}{N_{\text{ToA}}} \right)^{\frac{1}{2}} \mathcal{F}_{\sigma_{M_c}}(\cos i), \quad (18)$$

and

$$\text{Cov}[M_c, \sin i] = \left(\frac{M_\odot}{M_c} \right) \left(\frac{\sigma_{\text{rms}}}{\mu\text{s}} \right)^2 \left(\frac{1000}{N_{\text{ToA}}} \right) \mathcal{F}_{\text{Cov}[M_c, \sin i]}(\cos i). \quad (19)$$

We used our model to calculate σ_{M_p} for both PSRs J1614–2230 and J1909–3744. Because the ToA uncertainties range from 201 ns to $7.6 \mu\text{s}$ for PSR J1614–2230, and from 9 ns to $2.2 \mu\text{s}$ for PSR J1909–3744, we needed to create ToAs that are averaged in time. We averaged to 4 min (the observing length of typical observations), and then calculated the number of unique averaged ToAs and the weighted rms of the residuals. After averaging, for PSR J1614–2230 we obtained a weighted rms residual of $0.310 \mu\text{s}$ for 119 observations, and for PSR J1909–3744 we obtained a weighted rms residual of $0.125 \mu\text{s}$ for 179 observations. Using our model, we determined σ_{M_p} to be $\sim 0.03 M_\odot$ for both, which is in very good agreement with the measured σ_{M_p} of $0.03 M_\odot$ in Table 1.

We are able to use our model to estimate σ_{M_p} in PSRs J0101–6422 and J1514–4946, but not in PSRs J1101–6424 and J1732–5049, because our model is only reliable for $\cos i < 0.35$. For PSR J0101–6422 and J1514–4946, if we assume $M_p = 1.4 M_\odot$, by taking the median values of $\cos i$ to be 0.27 and 0.22, resulting in the companion masses to be 0.17 and $0.18 M_\odot$, we estimated σ_{M_p} to be 1.6 and $0.8 M_\odot$, respectively. If we assume $M_p = 2.0 M_\odot$, by taking the median values of $\cos i$ to be 0.30 and 0.27, resulting in the companion masses to be 0.21 and $0.23 M_\odot$, we estimated σ_{M_p} to be 2.0 and $1.3 M_\odot$, respectively.

4.2.2 Constant f_m

Soon after a binary pulsar is discovered, the mass function is quickly (and accurately) established, and so we also wanted to explore what

the likely σ_{M_p} would be if the pulsar mass and mass function are held fixed and we have no a priori knowledge of the inclination angle. This simulation method can be used whenever a new MSP is discovered, the mass function is known, and observers want to determine the chance of being able to measure a significant Shapiro delay for various inclination angles (and hence component masses). In this second set of simulations, we thus investigated how σ_{M_p} depended upon the known mass functions in PSRs J0101–6422, J1101–6424, J1514–4946, and J1732–5049 if $80^\circ \leq i < 90^\circ$, corresponding to $\sim 0.0 \leq \cos i \leq 0.16$. We did not explore higher $\cos i$ values as beyond this σ_{M_p} becomes larger than $2.0 M_\odot$. The companion mass values were calculated at different inclination angles using equation (11), assuming the mass function of each pulsar and $M_p = 1.5 M_\odot$. We then simulated 1300 ToAs with the rms residuals of data in Tables A1 and A2. The same procedure as Section 4.2.1 was used for performing the timing analysis, and plotting the errors. The only difference was, at every inclination angle, the simulation was repeated 100 times.

In Fig. 4, PSR J1101–6424 has the largest uncertainties in $\sin i$ and M_c due to its high rms residuals of $7.973 \mu\text{s}$ despite its massive companion. In our timing analysis for PSRs J0101–6422, J1101–6424, J1514–4946, and J1732–5049, we were interested in getting $\sigma_{M_p} \leq 0.1 M_\odot$ to be astrophysically interesting. However, from Fig. 4, the orbital inclination angles of PSRs J0101–6422 and J1514–4946 systems need to have $\cos i \leq 0.01$ to achieve this, given our timing precision. PSRs J1101–6424 and J1732–5049 have σ_{M_p} greater than $0.1 M_\odot$ at all values of $\cos i$. From the χ^2 analysis in 4.1, all four pulsars have $\cos i$ greater than 0.16, which stops us from placing meaningful constraints on their pulsar masses.

5 DISCUSSION

5.1 Comparison with previous measurements

We now check whether our results are consistent with those obtained by other authors, in a historical sequence.

5.1.1 PSR J1909–3744

From the NANOGrav 11-yr data release, Arzoumanian et al. (2018) measured the Shapiro delay using the traditional parameterisation (M_c , $\sin i$) for PSR J1909–3744 and estimated the parameters of $M_p = 1.48 \pm 0.03 M_\odot$, $M_c = 0.208 \pm 0.002 M_\odot$, and $i = 86^\circ 47' \pm 0^\circ 10'$. Liu et al. (2020) used 15 yr of PSR J1909–3744 observations with Nanca Radio Telescope to perform a high-precision analysis, and derived a pulsar mass of $M_p = 1.492 \pm 0.014 M_\odot$ and a companion mass of $M_c = 0.209 \pm 0.001 M_\odot$ using the traditional parameterization. Another independent analysis of J1909–3744 used 15 yr of PPTA-DR2e (Reardon et al. 2021) to measure the Shapiro delay parameters and obtained $M_p = 1.486 \pm 0.011 M_\odot$, $M_c = 0.2081 \pm 0.0009 M_\odot$, $i = 86^\circ 46' \pm 0^\circ 05'$. Using the International Pulsar Timing Array second data release (consisting of data sets collected by PPTA, the European pulsar timing array, and NANOGrav), Perera et al. (2019) analyzed 10.8 yr of data and provided measurements of the classical Shapiro parameters of $M_c = 0.209 \pm 0.001 M_\odot$ and $\sin i = 0.99807 \pm 0.00006$ for PSR J1909–3744. Using the mass function from `psrcat`⁷ (Manchester et al. 2005) and applying equations (11) and (13), we can derive the pulsar mass to be $M_p = 1.496 \pm 0.011 M_\odot$. Our derived masses and the inclination angle for PSR J1909–3744 (from Table 1) are

⁷<https://www.atnf.csiro.au/research/pulsar/psrcat>

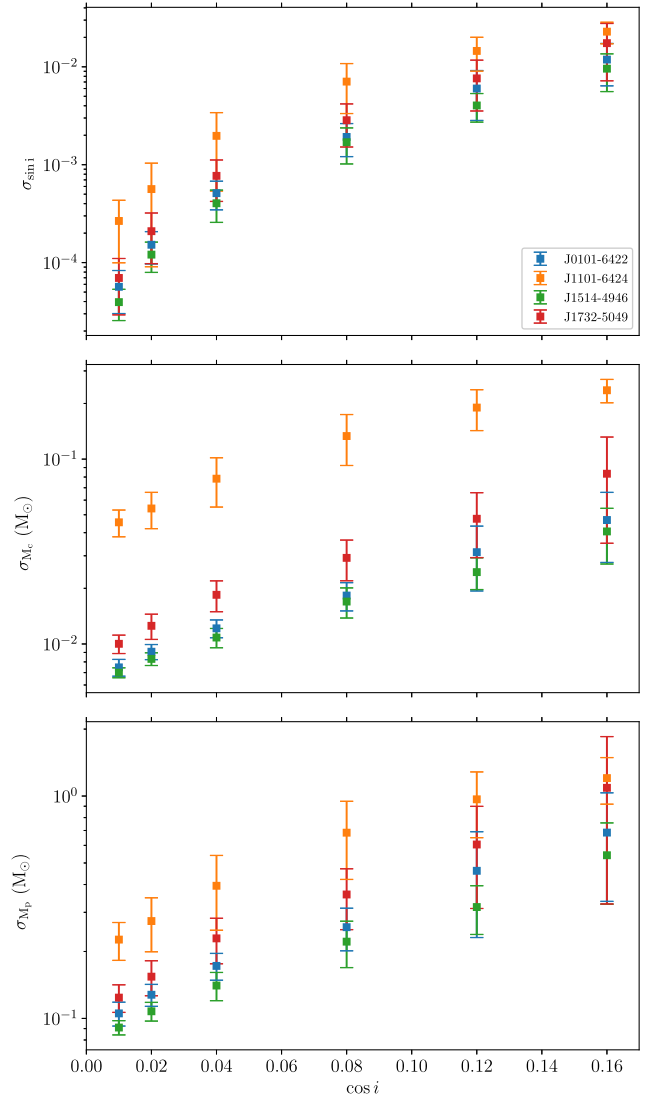


Figure 4. Error in the component masses at different orbital inclination angles for PSRs J0101–6422, J1101–6424, J1514–4946, and J1732–5049 using 1300 simulated ToAs with the rms of the actual data. Top panel: the error in the sine of orbital inclination angle. Middle panel: the error in the companion mass. Bottom panel: the error in the pulsar mass. The colours indicate the different binary pulsar systems of PSRs J0101–6422 (blue), J1101–6424 (orange), J1514–4946 (green), and J1732–5049 (red). The mass of the pulsar is assumed to be $1.5 M_\odot$ in all the simulations.

$M_p = 1.45 \pm 0.03 M_\odot$, $M_c = 0.205 \pm 0.003 M_\odot$, and $i = 86^\circ 69' \pm 0^\circ 10'$. Our masses are $\sim 1\sigma$ smaller than previously reported values.

5.1.2 PSR J1614–2230

The analysis of the PSR J1614–2230 system by Demorest et al. (2010) utilized a set of observations from the Green Bank Telescope to constrain the Shapiro delay using the classic parameterization: $M_p = 1.97 \pm 0.04 M_\odot$, $M_c = 0.500 \pm 0.006 M_\odot$, $i = 89^\circ 17' \pm 0^\circ 02'$. In a subsequent analysis, Fonseca et al. (2016) used the NANOGrav 9-yr data set, including a part of the data set used by Demorest et al. (2010), and made significant measurements of the Shapiro delay using both the classic and the orthometric parameterizations: $M_p = 1.928 \pm 0.017 M_\odot$, $M_c = 0.493 \pm 0.003 M_\odot$, $i = 89^\circ 189' \pm 0^\circ 014'$. Then Arzoumanian et al. (2018) used the NANOGrav 11-

yr data set and put constraints on the Shapiro delay using the orthometric parameterization obtaining: $M_p = 1.908 \pm 0.016 M_\odot$, $M_c = 0.493 \pm 0.003 M_\odot$, and $i = 89^\circ 204 \pm 0^\circ 014$.

Our reported masses and the inclination angle for PSR J1614–2230 (from Table 1) are $M_p = 1.94 \pm 0.03 M_\odot$, $M_c = 0.495 \pm 0.005 M_\odot$, and $i = 89^\circ 179 \pm 0^\circ 013$. These are the first independent measurement using MTPA data sets and are in good agreement with the previous measurements reported.

The discovery of PSR J1614–2230 is important for reasons beyond the implications for the study of the EOS of the dense matter: It was also important to establish that neutron stars can be born with large masses. Tauris, Langer & Kramer (2011) carried out a detailed study on modelling the formation and evolution of PSR J1614–2230 based on its physical parameters reported by Demorest et al. (2010). They investigated the evolution of PSR J1614–2230 using an evolving intermediate-mass X-ray binary and reported two possible scenarios: (1) through a common envelope started with their ‘Case C’ RLO (2) through the highly super-Eddington isotropic re-emission mode during their ‘Case A’ RLO, with the conclusion of the second scenario as the more likely one and the mass of the PSR J1614–2230 progenitor to be higher than $20 M_\odot$, and the initial companion to be a main-sequence donor star with the mass of $4.0\text{--}5.0 M_\odot$.

Because in Case A RLO the mass transfer happens while the companion is in the main sequence, the recycling episode is very long; providing a better explanation for the spin period of PSR J1614–2230, which is unusually small compared to all other pulsars with massive WDs. Interestingly, they find that even with the long accretion episode of Case A RLO, the total mass transfer is very small, less than $0.3 M_\odot$. This means that the birth mass was at least $\sim 1.7 M_\odot$. This is a strong indication that neutron star masses are generally acquired at birth, and don’t owe much to subsequent evolution.

Our measurements confirm their interpretation of the mass of PSR J1614–2230 system. The idea that neutron stars have high masses at birth was later confirmed with greater confidence by the discovery of massive neutron stars where little or no accretion happened, like PSR J2222–0137 (Cognard et al. 2017; Guo et al. 2021), and further by measurement of the mass of PSR J1933–6211, which is thought to have evolved from case A RLO and nevertheless has a low mass, confirming that not much mass was transferred during its long evolutionary episode (Geyer et al., submitted).

5.1.3 PSR J1125–6014

From the second data release of the Parkes Pulsar Timing Array (PPTA-DR2e), Reardon et al. (2021) performed an analysis of the PSR J1125–6014 system using more than 12 yr of data (a factor of four larger than the time span of PSR J1125–6014 observations with MeerKAT), and they constrained the pulsar mass, the companion mass, and the orbital inclination angle to be $M_p = 1.5 \pm 0.2 M_\odot$, $M_c = 0.31 \pm 0.03 M_\odot$, and $i = 77.9^{+1.4}_{-1.3}$, respectively. Our derived masses and the inclination angle for PSR J1125–6014 (from Table 1) are $M_p = 1.68^{+0.17}_{-0.15} M_\odot$, $M_c = 0.33 \pm 0.02 M_\odot$, and $i = 77.6^{+0.7}_{-0.8}$, which are in good agreement with Reardon et al. (2021).

The mass of the companion of PSR J1125–6014 is significantly higher than the prediction of Tauris & Savonije (1999) for the orbital period of this system ($0.25 M_\odot$). This suggests that the companion is not likely a Helium WD, but instead a low-mass CO WD. Interestingly, the short spin period of this pulsar and the high degree of recycling also suggest a case A RLO evolution, making the system a less massive version of PSR J1614–2230. We note in this

regard the similarity of the orbital periods (both ~ 8.7 d) and orbital eccentricities of the two systems.

5.2 Mass measurements with Shapiro delay

As an application of our error model, we can assess the fraction of MSPs for which we expect the masses to be measured to better than 10 per cent using the Shapiro delay method. For a sample of 100 randomly oriented MSPs, each with a pulsar mass of $1.4 M_\odot$, 1300 ToAs, and the rms residuals of $1 \mu\text{s}$, we would expect to measure ~ 11 MSP masses to better than 10 per cent, assuming the companion masses are all $0.2 M_\odot$. If we increase the companion mass to be $0.4 M_\odot$, we expect to measure up to 19 MSP masses to this accuracy. Regarding this, the maximum angles that we expect the systems to be inclined at are $\sim 84^\circ$ and 79° for the MSP systems with companion masses of 0.2 and $0.4 M_\odot$, respectively. Therefore, the fraction of the MSPs in a MSP sample for which we expect to measure their masses to this accuracy using the Shapiro delay method is low.

5.3 The mass distribution for WD companions to pulsars

For all known pulsars that are located outside of globular clusters and whose companions are found to be WDs, we have plotted the precisely measured masses (with the pulsar mass uncertainty less than 15 per cent), with logarithmic scaled coloured circles indicating the pulsar spin period, in the panel (a) of Fig. 5, as an update on figure 6 in McKee et al. (2020).

In that reference, they indicated two main populations in their plot: (1) systems with companions less massive than $0.46 M_\odot$ as low-mass helium WD stars (2) systems with companion masses between 0.7 and $0.9 M_\odot$ as carbon–oxygen WD stars. One of the conclusions is that there is a mass gap between 0.41 and $0.7 M_\odot$. In addition to these, there are five special systems with less common types of evolution.

First, in the case of PSR J1141–6545 and B2303+46, the WDs formed before the pulsar; this is the reason why these systems have eccentric orbits as their pulsars have not been recycled (van Kerkwijk & Kulkarni 1999; Tauris & Sennels 2000), and are therefore significantly slower (see panel b of Fig. 5). The remaining three systems have low-eccentricity systems and recycled pulsars. As discussed previously, PSR J1614–2230 is a rare system with a heavy neutron star and a carbon–oxygen WD, thought to have been formed in Case A RLO (Tauris et al. 2011). PSR J0348 + 0432 is a massive pulsar with a low-mass WD companion, has a tight orbit of 2.4 h, and the WD probably did not have an envelope mass less than the critical limit for hydrogen fusion, based on the assumption of Antoniadis et al. (2013). This pulsar is unusually slow for such a tight orbit and such a low-mass companion. Finally, PSR J2222–0137 has a very massive ($M_{\text{WD}} = 1.3194 \pm 0.0040 M_\odot$, Guo et al. 2021) and cool (Kaplan et al. 2014) WD companion.

We have explored where PSRs J1125–6014, J1614–2230, and J1909–3744 lie in Fig. 5 relative to the other pulsars. They are very consistent with the established distribution of the pulsars and WDs as discussed in McKee et al. (2020). PSR J1125–6014 is starting to fill an empty region of parameter space close to PSR J1614–2230, possibly because it also formed via Case A RLO.

Fig. 5 highlights the apparent gap between the masses of the MSPs with $0.15\text{--}0.5 M_\odot$ (this is even wider if we exclude binaries formed in Case A RLO), and those between 0.7 and $0.9 M_\odot$. Are there WD companions to neutron stars with masses in between? If not, this would suggest that we have a higher probability of measuring a

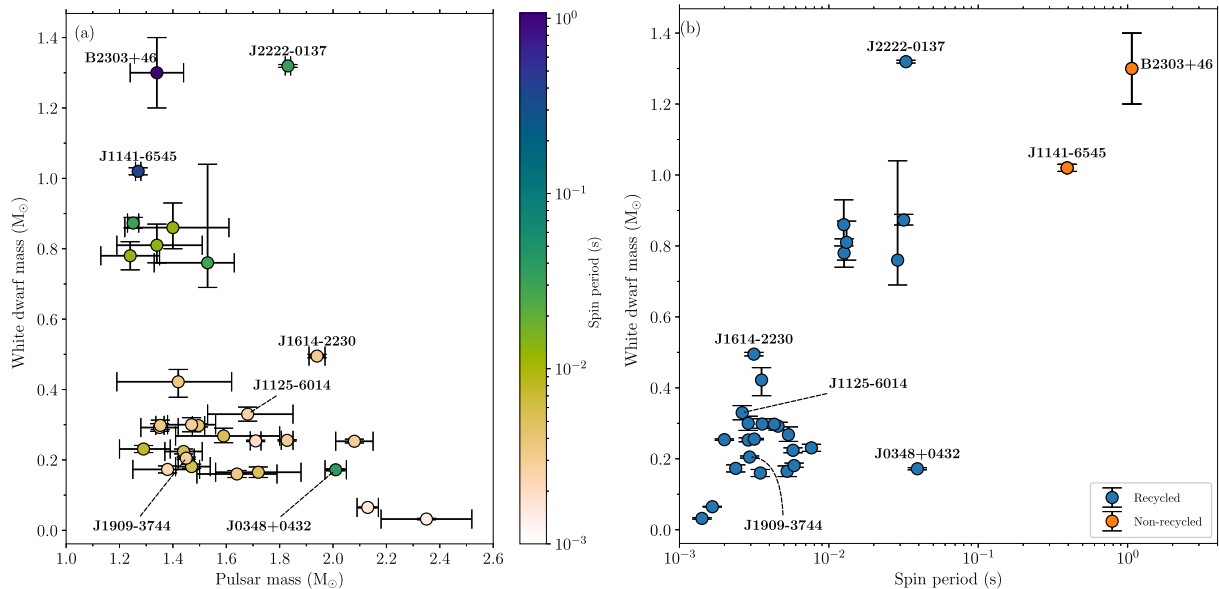


Figure 5. Precise mass measurements for all known pulsar-WD systems located outside of globular clusters. Panel (a): pulsar masses versus white dwarf (WD) companion masses with coloured circles indicating the spin period of pulsars. Panel (b): WD companion masses versus spin periods of pulsars. The values of the pulsar and WD masses, as well as their references, are listed in Table B1, and the spin periods are taken from `psrcat`.

Shapiro delay when the mass function yields $0.7 < M_c < 0.9 M_\odot$, assuming $i = 90^\circ$. However, it is important to attempt to measure the Shapiro delay in systems where the mass function suggests intermediate masses in order to test the idea that there is indeed a gap in WD masses between 0.4 and $0.7 M_\odot$.

In panel (b) of Fig. 5, we have also plotted the precisely measured companion mass versus pulsar spin period. For the bulk of the recycled pulsars, there is a strong correlation between the companion mass and the log of the spin period. This is expected from basic considerations of stellar evolution: The more massive WD companions had more massive progenitors, which have evolved faster. Because of this faster evolution, the time available for mass transfer and accretion is much smaller; resulting in smaller ablation of the B-field and a smaller amount of spin-up. The exceptions are PSR J0348+0242 and, as discussed before, PSR J1614–2230.

Sometimes extending the timing baseline with archival data can lead to increased accuracy of parameters that are undergoing secular changes, like position (via proper motion) and the time derivative of the projected semimajor axis. These can help limit the inclination angle from equation 12. We added 817 ToAs from 8.34 yr of PPTA DR2e data sets of PSR J1732–5049 (Reardon et al. 2021) with an rms residuals of $4.471 \mu\text{s}$, to the current 4486 MeerKAT ToAs with an rms residuals of $2.3 \mu\text{s}$ in order to try and constrain $\dot{x}$ and find an upper limit on inclination angle; however, by using 18.97 yr of data we could only gain an upper limit on the measured $\dot{x} = 4(7) \times 10^{-16}$, so we were still not be able to achieve this.

6 CONCLUSIONS

We have presented the timing analysis of seven MSPs using ~ 3 yr of MeerKAT observations, and provided updated timing models. We detected evidence for the relativistic Shapiro delay in the timing residuals of all MSPs. By conducting a Bayesian analysis, we were able to measure the Shapiro delay using the orthometric amplitude h_3 and orthometric ratio ζ . For PSRs J1125–6014, J1614–2230,

and J1909–3744, we could constrain the mass of the pulsars to be $M_p = 1.68^{+0.17}_{-0.15}$, 1.94 ± 0.03 , and $1.45 \pm 0.03 M_\odot$, and the companion masses to be $M_c = 0.33 \pm 0.02$, 0.495 ± 0.005 , and $0.205 \pm 0.003 M_\odot$, respectively. For pulsars with no meaningful mass constraints, 68 per cent likelihood of the companion masses and the orbital inclination angles were calculated using χ^2 analysis, assuming fixed pulsar masses of 1.2 and $2.0 M_\odot$. We find that our mass measurements are in agreement with previous mass measurements for the same systems, and with the previously observed distribution of pulsar and WD masses.

We performed two series of simulations for exploring the change of errors in the Shapiro parameters, $\sigma_{\sin i}$ and σ_{M_c} , at different orbital inclination angles, and derived the estimated errors in the pulsar mass σ_{M_p} . We then modelled the change of σ_{M_p} as a function of the cosine of the orbital inclination angle. This model works well for $\cos i < 0.35$. In addition, using our simulations, we demonstrated that, at the likely inclination angles of the four systems, it was no surprise that their pulsar masses could not be determined. We provided an empirical polynomial formula that observers can use to estimate their likelihood of determining pulsar masses given good orbital coverage and the number and rms of timing residuals as a function of the orbital inclination angle.

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DATA AVAILABILITY

Data is available from the Swinburne pulsar portal: <https://pulsars.or.g.au>.

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APPENDIX A: PULSAR EPHEMERIDES

Tables of pulsar observational and derived parameters. The standard error on the last quoted digit is indicated by the values in parentheses.

Table A1. Observed and derived timing model parameters for binary pulsars J0101–6422, J1101–6424, and J1125–6014.

Pulsar	PSR J0101–6422	PSR J1101–6424	PSR J1125–6014
General parameters			
Reference epoch (MJD)	59130	59132	59174
MJD range	58595–59666	58575–59689	58589–59758
Fit χ^2 /number of degrees of freedom	3018.65/3056	7574.58/7612	979.38/1027
Post-fit RMS of residuals (μ s)	1.602	7.973	0.336
L-band observation time (h)	14.37	39.17	11.59
UHF observation time (h)	11.84	–	–
Total number of observations L band/UHF	35/10	80/0	37/0
Spin and astrometric parameters			
Right ascension, α (epoch J2000; hh:mm:ss.s)	01:01:11.13637(5)	11:01:37.19156(4)	11:25:55.243407(4)
Declination, δ (epoch J2000; dd:mm:ss.s)	–64:22:30.2768(4)	–64:24:39.33311(17)	–60:14:06.81026(3)
Parallax, π (mas)	4.5(17)	–	1.6(2)
Proper motion in RA, $\mu_\alpha \cos \delta$ (mas yr $^{-1}$)	13.2(4)	–1.3(2)	11.07(3)
Proper motion in Dec., μ_δ (mas yr $^{-1}$)	–9.6(4)	–0.1(3)	–13.08(3)
Total proper motion, μ (mas yr $^{-1}$)	16.3(5)	1.4(3)	17.13(3)
Spin period, P (ms)	2.573 152 013 652 17(9)	5.109 272 984 057 81(12)	2.630 380 782 509 540(5)
First derivative of spin period, $\dot{P}$ (10^{-20} s s $^{-1}$)	0.5083(6)	0.2591(7)	0.37303(3)
Dispersion measure, DM (cm $^{-3}$ pc)	11.9217(4)	207.3647(5)	52.93371(6)
First derivative of DM, $\dot{DM}$ (cm $^{-3}$ pc s $^{-1}$)	0.0005(5)	–0.0009(3)	–0.00105(3)
Second derivative of DM, $\ddot{DM}$ (cm $^{-3}$ pc s $^{-2}$)	–0.0010(11)	0.0000(12)	0.00186(15)
Orbital parameters			
Orbital period, P_b (d)	1.787 596 7338(5)	9.611 708 8579(9)	8.752 603 650 59(15)
Projected semimajor axis of binary orbit, x (lt-s)	1.701 0463(2)	14.02463520(16)	8.339 198 93(3)
Time of ascending node, T_{asc} (MJD)	59100.476 969 27(7)	59081.94129419(3)	59167.119 295153(4)
First Laplace-Lagrange parameter, ϵ_1 (10^{-6})	–0.3(2)	21.57(3)	0.512(6)
Second Laplace-Lagrange parameter, ϵ_2 (10^{-6})	0.58(15)	3.94(3)	–0.612(5)
Orthometric amplitude, h_3 (μ s)	0.49(11)	0.32(12)	0.84(2)
Orthometric ratio, ζ	0.66(8)	0.81(15)	0.804(11)
Stochastic parameters			
Log $_{10}$ [EFAC] KAT receiver	0.008(10)	–0.010(9)	0.05(2)
Log $_{10}$ [EFAC] UHF receiver	0.030(9)	–	–
Log $_{10}$ [EQUAD] KAT receiver	–8.3(1.0)	–5.57(16)	–6.69(4)
Log $_{10}$ [EQUAD] UHF receiver	–7.9(1.2)	–	–
Log $_{10}$ [ECORR] KAT receiver	–8.5(9)	–8.5(9)	–8.4(8)
Log $_{10}$ [ECORR] UHF receiver	–8.8(8)	–	–
Log $_{10}$ [Red Amp]	–	–	–
Red Index	–	–	–
Log $_{10}$ [DM Amp]	–11.15(18)	–10.94(13)	–11.74(11)
DM Index	2.1(7)	2.8(7)	1.6(5)

Table A2. Observed and derived timing model parameters for binary pulsars J1514–4946, J1614–2230, J1732–5049, and J1909–3744.

Pulsar	PSR J1514–4946	PSR J1614–2230	PSR J1732–5049	PSR J1909–3744
General parameters				
Reference epoch (MJD)	59242	59099	59067	59059
MJD range	58834–59651	58589–59691	58526–59608	58526–59592
Fit χ^2 /number of degrees of freedom	461.05/463	1673.19/1740	4437.87/4467	3523.40/2936
Post-fit RMS of residuals (μ s)	1.487	0.648	2.300	0.090
<i>L</i> -band observation time (<i>h</i>)	11.59	7.02	18.79	23.66
UHF observation time (<i>h</i>)	—	—	2.84	—
Total number of observations <i>L</i> band/UHF	37/0	69/0	76/5	187/0
Spin and astrometric parameters				
Right ascension, α (epoch J2000; hh:mm:ss.s)	15:14:19.10584(4)	16:14: 36.50968(4)	17:32:47.766091(11)	19:09:47.424651(5)
Declination, δ (epoch J2000; dd:mm:ss.s)	−49:46:15.6363(4)	−22:30:31.539(3)	−50:49:00.3137(3)	−37:44:14.91316(19)
Parallax, π (mas)	—	1.11(17)	0.8(6)	0.75(8)
Proper motion in RA, $\mu_\alpha \cos \delta$ (mas yr ^{−1})	−7.8(2)	4.9(7)	−0.62(17)	−9.60(5)
Proper motion in Dec. μ_δ (mas yr ^{−1})	−10.3(4)	−26(4)	−9.6(4)	−35.6(2)
Total proper motion, (mas yr ^{−1})	13.0(4)	27(4)	9.6(5)	36.8(3)
Spin period, <i>P</i> (ms)	3.589 336 741 328 85(6)	3.150 807 657 99915(4)	5.312 550 296 151 66(7)	2.947108 074 685 310(15)
First derivative of spin period, $\dot{P}$ (10 ^{−20} s s ^{−1})	1.8670(10)	0.9625(4)	1.4182(5)	1.4023(3)
Dispersion measure, <i>DM</i> (cm ^{−3} pc)	31.008 53(13)	34.487 55(14)	56.821 89(13)	10.390 90(5)
First derivative of DM, $\dot{DM}$ (cm ^{−3} pc s ^{−1})	−0.000 16(15)	−0.000 54(8)	−0.00146(16)	0.000 053(12)
Second derivative of DM, $\ddot{DM}$ (cm ^{−3} pc s ^{−2})	0.0007(5)	−0.0003(3)	0.0009(4)	−0.00020(14)
Orbital parameters				
Orbital period, <i>P</i> _b (d)	1.922 653 5542(3)	8.686 619 4264(4)	5.2629972188(15)	1.533 449 477 320(12)
Projected semimajor axis of binary orbit, <i>x</i> (lt-s)	1.933 2622(3)	11.291 207 02(7)	3.982 869 55(19)	1.897994789(14)
Time of ascending node, <i>T</i> _{asc} (MJD)	59211.985 350 52(7)	59098.046 639 722(11)	59064.553 069 32(4)	59057.600 9102 99(3)
First Laplace-Lagrange parameter, ϵ_1 (10 ^{−6})	4.0(3)	0.930(12)	1.65(9)	1.940(14)
Second Laplace-Lagrange parameter, ϵ_2 (10 ^{−6})	3.1(3)	−1.297(17)	−8.02(6)	−0.104(14)
Orthometric amplitude, <i>h</i> ₃ (μ s)	0.5(2)	2.340(19)	0.53(13)	0.845(7)
Orthometric ratio, ζ	0.81(13)	0.9858(3)	0.4(3)	0.9441(16)
Stochastic parameters				
Log ₁₀ [EFAC] KAT receiver	−0.044(15)	−0.003(8)	0.008(8)	0.010(7)
Log ₁₀ [EFAC] UHF receiver	—	—	−0.070(14)	—
Log ₁₀ [EQUAD] KAT receiver	−8.3(1.0)	−8.6(9)	−6.00(3)	−7.75(8)
Log ₁₀ [EQUAD] UHF receiver	—	—	−8.2(1.1)	—
Log ₁₀ [ECORR] KAT receiver	−8.7(8)	−7.7(5)	−8.9(7)	−7.57(4)
Log ₁₀ [ECORR] UHF receiver	—	—	−8.6(9)	—
Log ₁₀ [Red Amp]	—	−13.3(4)	—	−13.19(15)
Red Index	—	3.9(1.3)	—	2.6(6)
Log ₁₀ [DM Amp]	—	−11.41(11)	−11.39(9)	−11.83(9)
DM Index	—	1.4(4)	0.16(14)	1.8(3)

APPENDIX B: PULSAR AND COMPANION MASS MEASUREMENTS WITH THE RELEVANT REFERENCES

Table B1. Pulsars with estimated WD companion masses plotted in Fig. 5.

Pulsar	M_p	M_c	Reference
J0348+0432	$2.01^{+0.04}_{-0.04}$	$0.172^{+0.003}_{-0.003}$	Antoniadis et al. (2013)
J0437–4715	$1.44^{+0.07}_{-0.07}$	$0.224^{+0.007}_{-0.007}$	Reardon et al. (2016)
J0621+1002	$1.53^{+0.1}_{-0.2}$	$0.76^{+0.28}_{-0.07}$	Kasian (2012)
J0740+6620	$2.08^{+0.07}_{-0.07}$	$0.253^{+0.006}_{-0.005}$	Fonseca et al. (2021)
J0751+1807	$1.64^{+0.15}_{-0.15}$	$0.16^{+0.01}_{-0.01}$	Desvignes et al. (2016)
J0952–0607	$2.35^{+0.17}_{-0.17}$	$0.032^{+0.002}_{-0.002}$	Romani et al. (2022)
J0955–6150	$1.71^{+0.02}_{-0.02}$	$0.254^{+0.002}_{-0.002}$	Serylak et al. (2022)
J1012+5307	$1.72^{+0.16}_{-0.16}$	$0.165^{+0.015}_{-0.015}$	Mata Sánchez et al. (2020)
J1125–6014	$1.68^{+0.17}_{-0.15}$	$0.33^{+0.02}_{-0.02}$	This work
J1141–6545	$1.27^{+0.01}_{-0.01}$	$1.02^{+0.01}_{-0.01}$	Bhat, Bailes and Verbiest (2008)
J1614–2230	$1.94^{+0.03}_{-0.03}$	$0.495^{+0.005}_{-0.005}$	This work
J1713+0747	$1.35^{+0.07}_{-0.07}$	$0.292^{+0.011}_{-0.011}$	Arzoumanian et al. (2018)
J1738+0333	$1.47^{+0.07}_{-0.06}$	$0.181^{+0.007}_{-0.005}$	Antoniadis et al. (2012)
J1802–2124	$1.24^{+0.11}_{-0.11}$	$0.78^{+0.04}_{-0.04}$	Ferdman et al. (2010)
J1810+1744	$2.13^{+0.04}_{-0.04}$	$0.065^{+0.001}_{-0.001}$	Romani et al. (2021)
B1855+09	$1.37^{+0.13}_{-0.10}$	$0.244^{+0.014}_{-0.012}$	Arzoumanian et al. (2018)
J1909–3744	$1.45^{+0.03}_{-0.03}$	$0.205^{+0.003}_{-0.003}$	This work
J1918–0642	$1.29^{+0.1}_{-0.09}$	$0.231^{+0.010}_{-0.010}$	Arzoumanian et al. (2018)
J1933–6211	$1.42^{+0.20}_{-0.23}$	$0.422^{+0.035}_{-0.044}$	Geyer et al. 2022 (submitted)
J1946+3417	$1.828^{+0.022}_{-0.022}$	$0.2556^{+0.0019}_{-0.0019}$	Barr et al. (2017)
J1949+3106	$1.34^{+0.17}_{-0.15}$	$0.81^{+0.06}_{-0.05}$	Zhu et al. (2019)
J1950+2414	$1.496^{+0.023}_{-0.023}$	$0.2975^{+0.0046}_{-0.0038}$	Zhu et al. (2019)
J2043+1711	$1.38^{+0.12}_{-0.13}$	$0.173^{+0.010}_{-0.010}$	Arzoumanian et al. (2018)
J2045+3633	$1.251^{+0.021}_{-0.021}$	$0.873^{+0.016}_{-0.014}$	McKee et al. (2020)
J2053+4650	$1.40^{+0.21}_{-0.18}$	$0.86^{+0.07}_{-0.06}$	Berezina et al. (2017)
J2222–0137	$1.831^{+0.010}_{-0.010}$	$1.3194^{+0.0040}_{-0.0040}$	Guo et al. (2021)
J2234+0611	$1.353^{+0.014}_{-0.017}$	$0.298^{+0.015}_{-0.012}$	Stovall et al. (2019)
B2303+46	$1.34^{+0.10}_{-0.10}$	$1.30^{+0.10}_{-0.10}$	Thorsett & Chakrabarty (1999)
J2339–0533	$1.47^{+0.09}_{-0.09}$	$0.30^{+0.02}_{-0.02}$	Kandel et al. (2020)

APPENDIX C:

Posterior probability distributions of parameters. The figures are created using `corner.py` developed by Foreman-Mackey (2016).

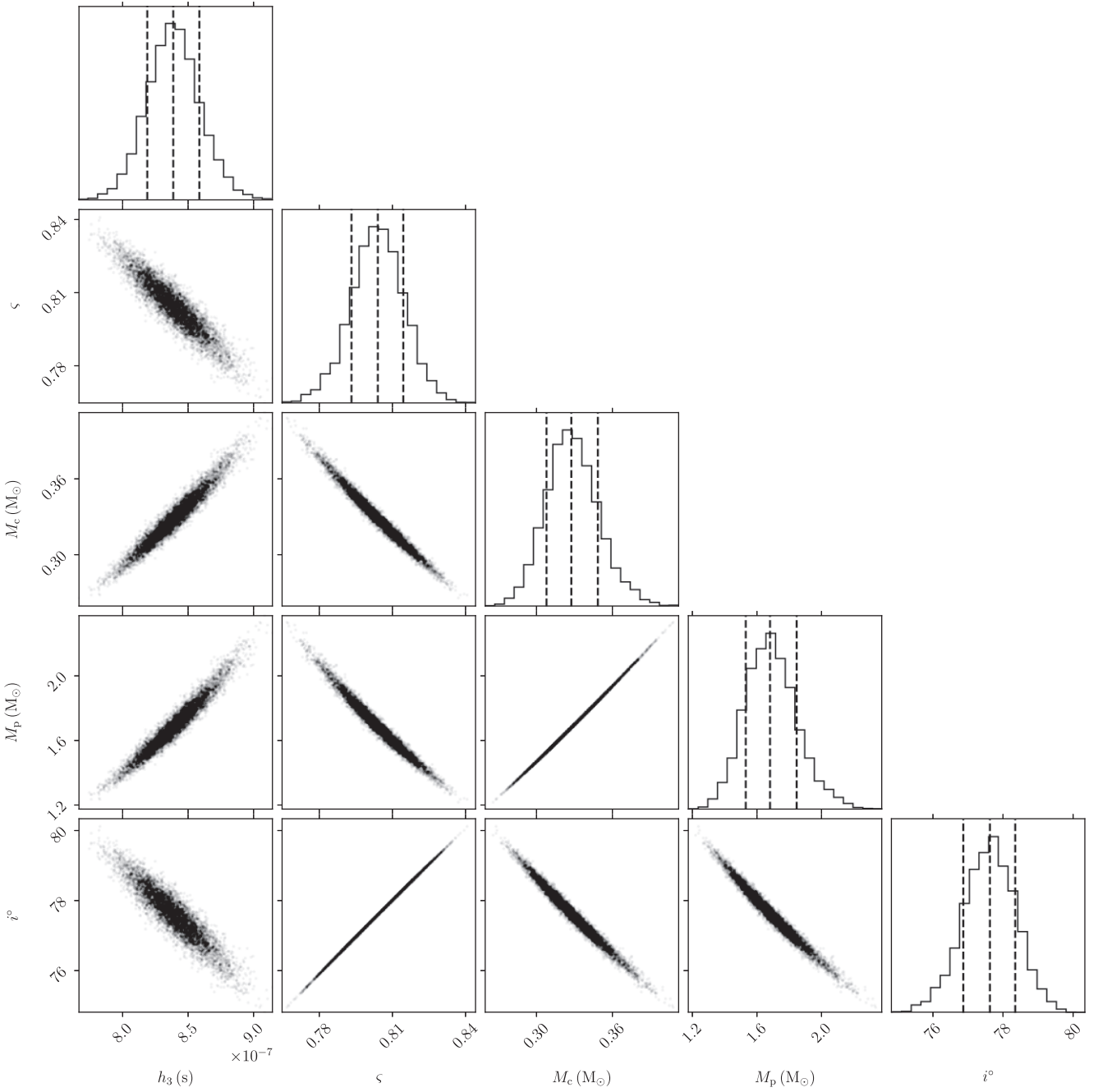


Figure C1. 2D marginalized posterior probability distributions of the timing parameters: h_3 and ζ , and the derived parameters: M_c , M_p , i , for J1125–6014. The dashed vertical lines on the 1D histograms are showing the median and 68 per cent confidence levels.

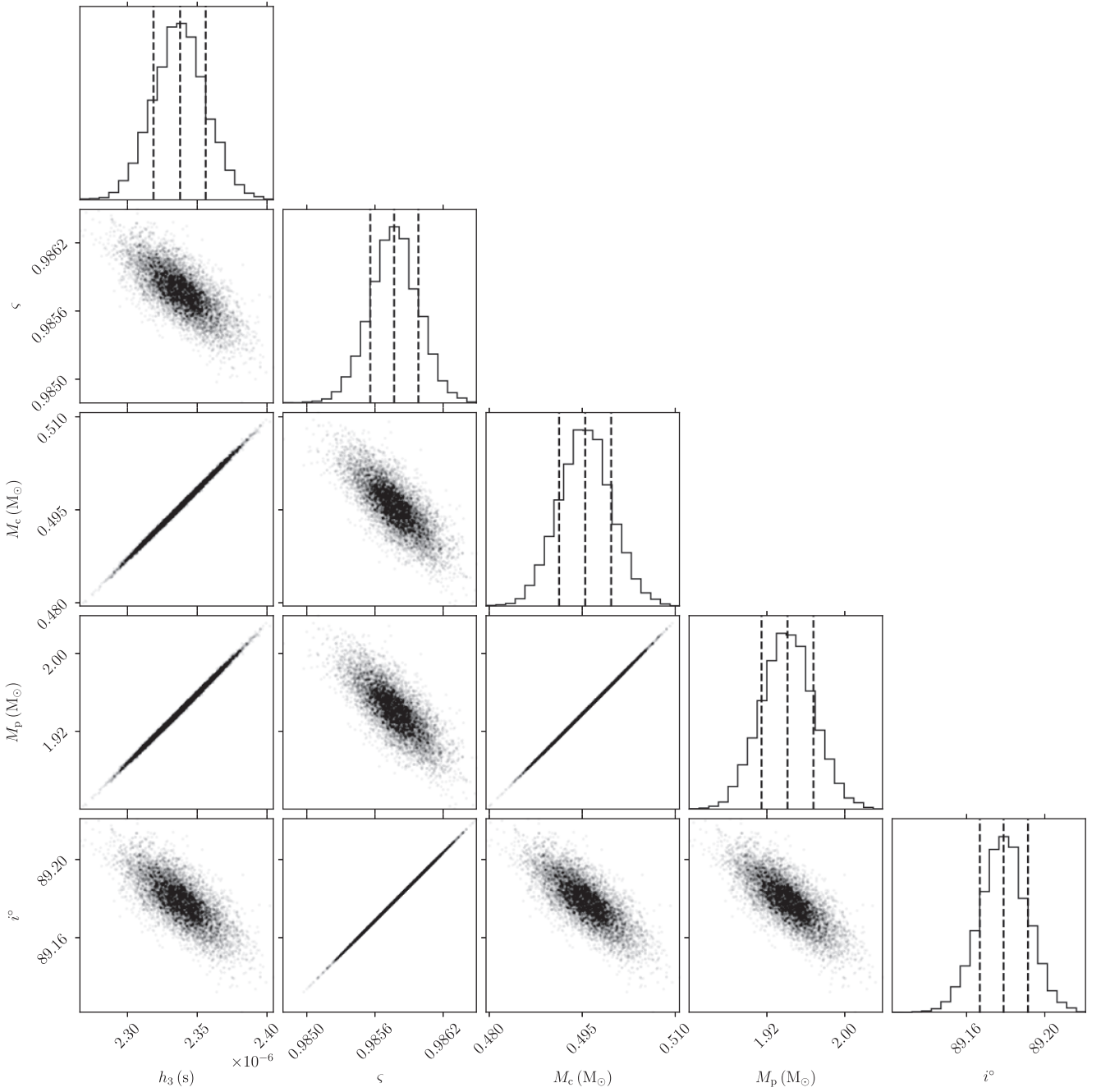


Figure C2. 2D marginalized posterior probability distributions of the timing parameters: h_3 and ζ , and the derived parameters: M_c , M_p , i , for J1614–2230. The dashed vertical on the 1D histograms lines are showing the median and 68 per cent confidence levels..

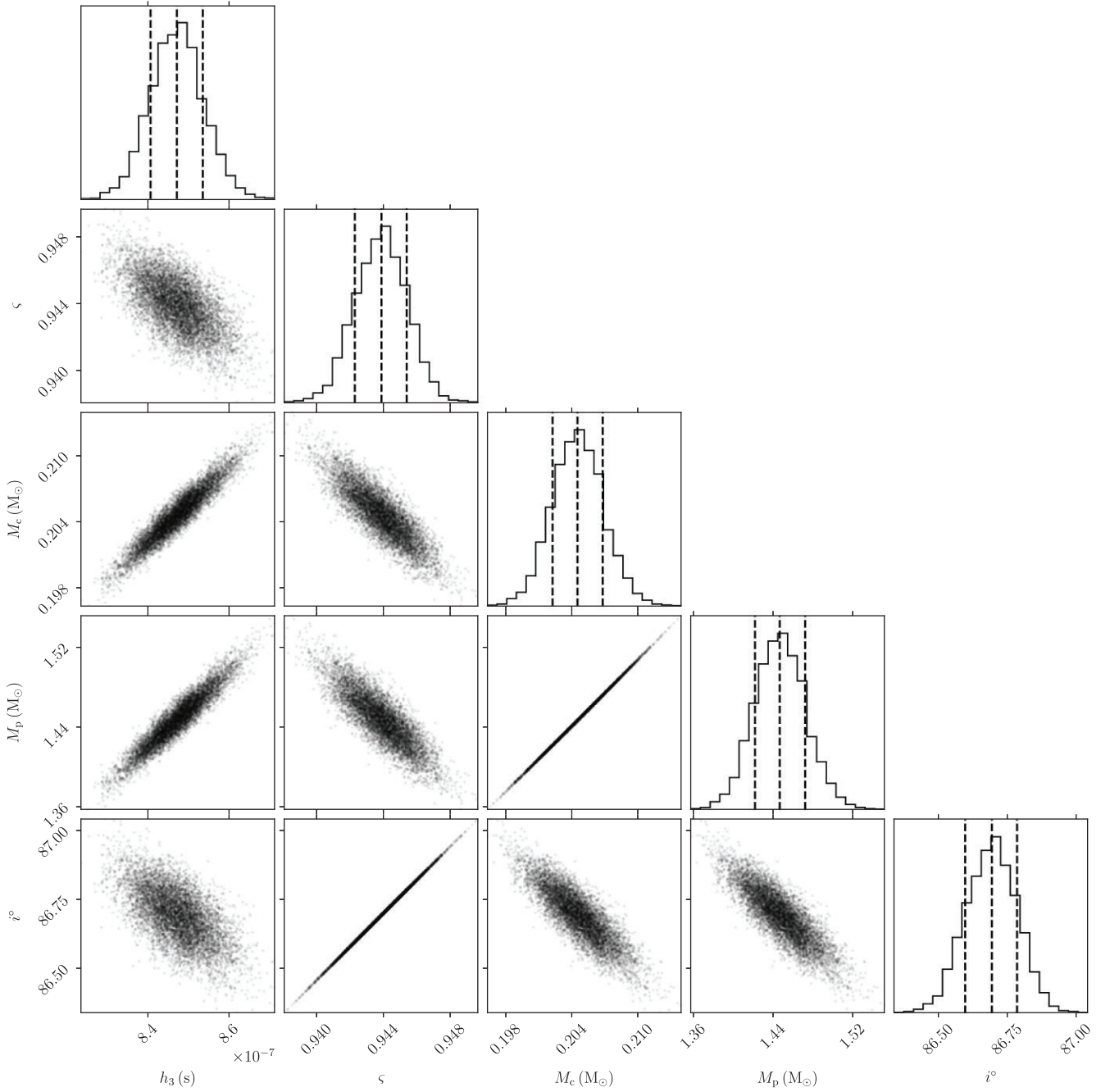


Figure C3. 2D marginalized posterior probability distributions of the timing parameters: h_3 and ζ , and the derived parameters: M_c , M_p , i , for J1909–3744. The dashed vertical lines on the 1D histograms are showing the median and 68 per cent confidence levels.

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